

TMF630 REST API Design Guidelines Part 5: JSON Patch extensions to manage arrays

TMF630 Part 5

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Executive Summary

The “REST API Design Guidelines” document provides guidelines and design patterns used in designing TM Forum REST APIs. The document is organized in multiple parts as follows:

- **Part One:** Practical guidelines for RESTful APIs; naming, CRUD, filtering, tasks, history, notifications
- **Part Two:** Advanced guidelines for RESTful API polymorphism, extension patterns, depth and expand directive, entity RefOrValue
- **Part Three:** Guidelines for extending TMF Open APIs with hypermedia support
- **Part Four:** Advanced guidelines for RESTful APIs; lifecycle management, common tasks
- **Part Five:** JSON Patch extension to manage arrays
- **Part Six:** JSON Path extension
- **Part Seven:** JSON Schema patterns
- **Part Eight:** API Specialization

This document, “REST API Design Guidelines Part 5” provides information for the development of TM Forum APIs and recommendations, guidelines on JSON Patch extension to manage arrays.

Conventions

Keywords to Indicate Requirement Levels

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in [\[RFC8174\]](#).

Formatting Conventions for Examples

In most examples, attributes occur that normally would be long UUID values; as simplification of the layout (avoidance of unnecessary line breaks), these values are shortened to a length of 4 characters only, followed by two dots:

`bd69..` instead of `bd694f23-627f-4553-a1df-7e79d62001ac`

For the same reason, in examples we abbreviate the leading part of URLs by a placeholder:

`{apiRoot}` instead of `https://mycsp.com/tmf-api`

To increase readability, we do not show the encoded form of URLs, but use the unencoded form. Of course, in a real implementation the URLs must be encoded as per [\[RFC3986\]](#).

Chapter 1. Introduction

JavaScript Object Notation (JSON) [\[RFC4627\]](#) is a common format for the exchange and storage of structured data, consisting of simple type data attributes, complex objects, and arrays.

HTTP PATCH [\[RFC5789\]](#) extends the Hypertext Transfer Protocol (HTTP) [\[RFC9110\]](#) with a method to perform partial modifications to resources.

JSON Patch [\[RFC6902\]](#) is a format for expressing a sequence of operations to apply to a target JSON document for use with the HTTP PATCH method.

The application of JSON Patch for making partial updates to a JSON data structure involves specifying the position of the affected element within the JSON data using a JSON Pointer, specified in [\[RFC6901\]](#). To identify the position of an element within an array, JSON Pointer uses the array's zero-based index of the element (note that even if the element has an `id` attribute, it is still the array index that a JSON Pointer uses as reference, not the `id`). Identification of an element without knowledge of its current position in the array is not possible, and further problems arise when elements are removed or added, changing the target element's index.

JSON Patch Query is a format that extends the functionality of JSON Patch, allowing for partial updates of array elements without requiring prior knowledge of their order within the array. It eliminates the need to refer to array indexes. Instead, the structure's various parameters serve as references to identify individual elements within the array.

Chapter 2. Structure of a JSON Patch Query Document

A JSON Patch Query document follows the same structure as a JSON Patch [\[RFC6902\]](#) document, which represents an array of objects, each object representing a single operation to be applied to the target JSON document, but the "path" member which in JSON Patch is only a JSON Pointer now includes means to allow identifying one or more elements within an array that are impacted by the operation.

Due to the additional functionality provided by JSON Patch Query, a new mime type that identifies a JSON structure as JSON Patch document is defined as:

```
application/json-patch-query+json
```

JSON Patch Query supports three mechanisms in which a query/condition can be passed to identify the target element(s) of the operation:

- extend the JSON Pointer in the **path** attribute by *query expressions*, as defined in part 1 section 4.4
- extend the JSON Pointer in the **path** attribute by *the filter parameter containing a JSON Path as value*, as defined in part 6
- replace the JSON Pointer in the **path** attribute by *a JSONPath expression*



It is expected that at least one of these options is supported by an API implementation. The preferred approach is to utilize the third option because of its standardized JSON Path syntax and wide libraries support.



With each of the above options, the path may identify a single target element, several target elements, or none at all. The operation applies to all target elements identified by the path.

Evaluation of a JSON Patch Query document is performed in the same way as for a JSON Patch document.

Chapter 3. Operations

Operations in a JSON Patch Query document are handled in the same way as for a JSON Patch document, but the query expression (query parameters, filter with JSON Path expression, or JSON Path expression) in the **path** member is used to identify the elements within a JSON structure that are impacted by the operation, the ones that match all the criteria included in the query expression.

When the query is not required because the array index can be used, this format is handled in the same way as JSON Patch [\[RFC6902\]](#).

Refer to Examples section below for a list of examples of the use of JSON Patch Query document for different operations on a resource.

Chapter 4. Error Handling

Same as for a JSON Patch document, if a normative requirement is violated by a JSON Patch Query document, or if an operation is not successful on any of its identified target elements, evaluation of the entire document SHOULD terminate and application of the entire patch document SHALL NOT be deemed successful.

See [\[RFC5789\]](#), section 2.2 for considerations regarding handling errors when JSON Patch is used with the HTTP PATCH method, including suggested status codes to use to indicate various conditions.

In addition to the above, if a JSON Path expression is being used (part of the **filter** selector or not) and the expression fails at the syntax level a **400 Bad Request** (see [\[RFC9110\]](#), section 15.5.1) MUST be returned.

If the JSON Path is not supported by the server implementation, but the client is making use of it a **501 Not Implemented** (see [\[RFC9110\]](#) section-16.6.2) MUST be returned.

Chapter 5. JSON Path Expression

JSON Patch Query allows the incorporation of a JSON Path expression into a JSON Patch operation, enabling the identification of elements within arrays without reliance on the array position.

When using a JSON Path Expression this can be achieved in two ways:

- through the query syntax and using the **filter** selector (the JSON Path **filter** selector is documented in [Design Guidelines Part 6](#))
- passing directly the JSON Path expression



Passing the JSON Path expression is most likely the common way, however due to a limitation on how JSON Path traverses the document whilst performing a document query (in some complex scenarios) using the **filter** selector overcomes this issue.

A simple JSON Patch Query document where a JSON Path expression has been used as part of the **path** is as following:

```
[
  {
    "op": "replace",
    "path": "$.note[?(@.id=='5e74..')].author",
    "value": "Sandy Smith"
  }
]
```



Further examples demonstrating the intent, before and after update resource definition can be found below.

The following section covers the common operations using a set of examples of different **PATCH** update using JSON Patch Query document.

Considering the following *Trouble Ticket* resource (**64f1..**):

```
{
  "id": "64f1..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/64f1..",
  "creationDate": "2024-01-31T07:34:45.968Z",
  "description": "Reporting a fixed line outage. Please investigate and resolve"
```

```
as soon as possible.",
  "expectedResolutionDate": "2019-06-10T07:34:45.968Z",
  "lastUpdate": "2024-02-10T07:34:45.968Z",
  "name": "Fixed line outage",
  "priority": "High",
  "requestedResolutionDate": "2019-05-31T07:34:45.968Z",
  "severity": "Major",
  "status": "pending",
  "statusChangeDate": "2019-05-31T07:34:45.968Z",
  "statusChangeReason": "Need more information from the customer",
  "ticketType": "Outage",
  "attachment": [
    {
      "id": "2384..",
      "attachmentType": "screenshot",
      "mimeType": "image/png",
      "name": "CPE Diagnostic Output",
      "url": "https://mycsp.com/docloader?docnum=3534555",
      "@type": "AttachmentRef"
    }
  ],
  "channel": {
    "id": "4a62..",
    "name": "self service",
    "@type": "ChannelRef"
  },
  "note": [
    {
      "id": "5e74..",
      "author": "Jack Smith",
      "date": "2024-02-10T14:34:45.968Z",
      "@type": "Note"
    },
    {
      "id": "a7f2..",
      "author": "Emily Johnson",
      "date": "2024-02-01T11:34:45.968Z",
      "text": "Urgent matter, contact me at your earliest convenience",
      "@type": "Note"
    },
    {
      "id": "9b8a..",
      "author": "Support Team",
      "date": "2024-01-31T07:40:45.968Z",
      "text": "Urgent: We are currently experiencing a fixed line outage. Please
```

```
address this issue immediately.",
  "@type": "Note"
},
],
"relatedParty": [
  {
    "role": "member",
    "partyOrPartyRole": {
      "id": "eb10..",
      "href": "{apiRoot}/partyManagement/v5/individual/eb10..",
      "name": "Sandy Smith",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  },
  {
    "role": "reporter",
    "partyOrPartyRole": {
      "id": "387d..",
      "name": "Jacob Jac Miller",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  },
  {
    "role": "customer",
    "partyOrPartyRole": {
      "id": "b811..",
      "href": "{apiRoot}/partyManagement/v5/customer/b811..",
      "name": "Jack Smith",
      "@type": "PartyRoleRef",
      "@referredType": "Customer"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  }
],
"@type": "TroubleTicket"
}
```

5.1. Adding an attribute to one of the components from an array

Considering the resource example defined in [\[troubleTicketExample\]](#).

Add to the **attachment** array, for the element where the condition **id=2384..** evaluates to true, a new attribute **description** with a value of **CPE Output**

5.1.1. Using JSON Path

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..  
Content-Type: application/json-patch-query+json
```

```
[  
  {  
    "op": "add",  
    "path": "$.attachment[?(@.id=='2384..')].description",  
    "value": "CPE Output"  
  }  
]
```



The above JSON Path expression will resolve to **['attachment'][0]['description']**



As the JSON Patch operation is **add**, the last attribute from the JSON Path expression is pointing to the new attribute to be added.

or

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..  
Content-Type: application/json-patch-query+json
```

```
[  
  {  
    "op": "add",  
    "path": "$.attachment[?(@.id=='2384..')]",  
    "value": {"description": "CPE Output"}  
  }  
]
```



The above JSON Path expression will resolve to `$['attachment'][0]`

or

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..
Content-Type: application/json-patch-query+json
```

```
[
  {
    "op": "add",
    "path": "$.attachment[?(@.name=='CPE Diagnostic Output' &&
@.attachmentType=='screenshot')]",
    "value": {"description":"CPE Output"}
  }
]
```



The above JSON Path expression will resolve to `$['attachment'][0]`

All the above JSON Patch examples have the following meaning:

- add to the `attachment` array, for the element where the `id=2384..` a new attribute `description` with the value provided.

5.1.2. Using JSON Path filter selector

When using the `filter` selector, the query expression syntax is being used, with the first part passing the path (using JSON Pointer) and the second part containing the condition that needs to be evaluated.

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..
Content-Type: application/json-patch-query+json
```

```
[
  {
    "op": "add",
    "path": "/attachment/description?filter=attachment[?(@.id=='2384..')]",
  }
]
```

```

    "value": "CPE Output"
  }
]

```



As per [Design Guidelines – Part 6](#), for the **filter** selector the **\$** can be omitted.



As the **filter** selector is being used the path to the new element is following the JSON Pointer notation.

All the above would result in the following updated Trouble Ticket resource:

RESPONSE

```

200 OK
Content-Type: application/json

```

```

{
  "id": "64f1..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/64f1..",
  "creationDate": "2024-01-31T07:34:45.968Z",
  "description": "Reporting a fixed line outage. Please investigate and resolve
as soon as possible.",
  "expectedResolutionDate": "2019-06-10T07:34:45.968Z",
  "lastUpdate": "2024-02-10T07:34:45.968Z",
  "name": "Fixed line outage",
  "priority": "High",
  "requestedResolutionDate": "2019-05-31T07:34:45.968Z",
  "severity": "Major",
  "status": "pending",
  "statusChangeDate": "2019-05-31T07:34:45.968Z",
  "statusChangeReason": "Need more information from the customer",
  "ticketType": "Outage",
  "attachment": [
    {
      "id": "2384..",
      "attachmentType": "screenshot",
      "description": "CPE Output",
      "mimeType": "image/png",
      "name": "CPE Diagnostic Output",
      "url": "https://mycsp.com/docloader?docnum=3534555",
    }
  ]
}

```

```

        "@type": "AttachmentRef"
    },
],
"channel": {
    "id": "4a62..",
    "name": "self service",
    "@type": "ChannelRef"
},
"note": [
    {
        "id": "5e74..",
        "author": "Jack Smith",
        "date": "2024-02-10T14:34:45.968Z",
        "text": "This is quite important, please approach me as soon as possible",
        "@type": "Note"
    },
    {
        "id": "a7f2..",
        "author": "Emily Johnson",
        "date": "2024-02-01T11:34:45.968Z",
        "text": "Urgent matter, contact me at your earliest convenience",
        "@type": "Note"
    },
    {
        "id": "9b8a..",
        "author": "Support Team",
        "date": "2024-01-31T07:40:45.968Z",
        "text": "Urgent: We are currently experiencing a fixed line outage. Please
address this issue immediately.",
        "@type": "Note"
    }
],
"relatedParty": [
    {
        "role": "member",
        "partyOrPartyRole": {
            "id": "eb10..",
            "href": "{apiRoot}/partyManagement/v5/individual/eb10..",
            "name": "Sandy Smith",
            "@type": "PartyRef",
            "@referredType": "Individual"
        },
        "@type": "RelatedPartyRefOrPartyRoleRef"
    },
    {

```

```

    "role": "reporter",
    "partyOrPartyRole": {
      "id": "387d..",
      "name": "Jacob Jac Miller",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  },
  {
    "role": "customer",
    "partyOrPartyRole": {
      "id": "b811..",
      "href": "{apiRoot}/partyManagement/v5/customer/b811..",
      "name": "Jack Smith",
      "@type": "PartyRoleRef",
      "@referredType": "Customer"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  }
],
"@type": "TroubleTicket"
}

```

5.2. Removing one of the components of an Array Element (the whole structure)

Considering the resource example defined in [\[troubleTicketExample\]](#).

Remove from the **note** array, the element identified by the path provided, the element/structure whose attribute **author** has a value of **Jack Smith**.

5.2.1. Using the JSON Path

REQUEST

```

PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..
Content-Type: application/json-patch-query+json

```

```

[
  {
    "op": "remove",

```

```
"path": "$.note[?(@.author=='Jack Smith')]"
}
```



The above JSON Path expression will resolve to `$['note'][0]`

5.2.2. Using JSON Path filter selector

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..
Content-Type: application/json-patch-query+json
```

```
[
  {
    "op": "remove",
    "path": "?filter=note[?(@.author=='Jack Smith')]"
  }
]
```



The above JSON Path expression will resolve to `$['note'][0]`



As per [Design Guidelines – Part 6](#), for the **filter** selector the `$` can be omitted.

All the above would result in the following updated Trouble Ticket resource:

RESPONSE

```
200 OK
Content-Type: application/json
```

```
{
  "id": "64f1..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/64f1..",
  "creationDate": "2024-01-31T07:34:45.968Z",
  "description": "Reporting a fixed line outage. Please investigate and resolve as soon as possible.",
}
```

```
"expectedResolutionDate": "2019-06-10T07:34:45.968Z",
"lastUpdate": "2024-02-10T07:34:45.968Z",
"name": "Fixed line outage",
"priority": "High",
"requestedResolutionDate": "2019-05-31T07:34:45.968Z",
"severity": "Major",
"status": "pending",
"statusChangeDate": "2019-05-31T07:34:45.968Z",
"statusChangeReason": "Need more information from the customer",
"ticketType": "Outage",
"attachment": [
  {
    "id": "2384..",
    "attachmentType": "screenshot",
    "mimeType": "image/png",
    "name": "CPE Diagnostic Output",
    "url": "https://mycsp.com/docloader?docnum=3534555",
    "@type": "AttachmentRef"
  }
],
"channel": {
  "id": "4a62..",
  "name": "self service",
  "@type": "ChannelRef"
},
"note": [
  {
    "id": "a7f2..",
    "author": "Emily Johnson",
    "date": "2024-02-01T11:34:45.968Z",
    "text": "Urgent matter, contact me at your earliest convenience",
    "@type": "Note"
  },
  {
    "id": "9b8a..",
    "author": "Support Team",
    "date": "2024-01-31T07:40:45.968Z",
    "text": "Urgent: We are currently experiencing a fixed line outage. Please address this issue immediately.",
    "@type": "Note"
  }
],
"relatedParty": [
  {
    "role": "member",
```

```

    "partyOrPartyRole": {
      "id": "eb10..",
      "href": "{apiRoot}/partyManagement/v5/individual/eb10..",
      "name": "Sandy Smith",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  },
  {
    "role": "reporter",
    "partyOrPartyRole": {
      "id": "387d..",
      "name": "Jacob Jac Miller",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  },
  {
    "role": "customer",
    "partyOrPartyRole": {
      "id": "b811..",
      "href": "{apiRoot}/partyManagement/v5/customer/b811..",
      "name": "Jack Smith",
      "@type": "PartyRoleRef",
      "@referredType": "Customer"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  }
],
"@type": "TroubleTicket"
}

```

5.3. Removing an attribute from one of the components of an array

Considering the resource example defined in [\[troubleTicketExample\]](#).

Remove from the **attachment** array, the attribute from the element/structure identified by the path provided, where the condition **id=2384..** is true

5.3.1. Using the JSON Path

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..  
Content-Type: application/json-patch-query+json
```

```
[  
  {  
    "op": "remove",  
    "path": "$.attachment[?(@.id=='2384..')].attachmentType"  
  }  
]
```



The above JSON Path expression will resolve to `$['attachment'][0]['attachmentType']`

5.3.2. Using JSON Path filter selector

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..  
Content-Type: application/json-patch-query+json
```

```
[  
  {  
    "op": "remove",  
    "path": "?filter=attachment[?(@.id=='2384..')].attachmentType"  
  }  
]
```



As per [Design Guidelines – Part 6](#), for the **filter** selector the `$` can be omitted.

All the above would result in the following updated Trouble Ticket resource:

RESPONSE

```
200 OK  
Content-Type: application/json
```

```
{
  "id": "64f1..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/64f1..",
  "creationDate": "2024-01-31T07:34:45.968Z",
  "description": "Reporting a fixed line outage. Please investigate and resolve
as soon as possible.",
  "expectedResolutionDate": "2019-06-10T07:34:45.968Z",
  "lastUpdate": "2024-02-10T07:34:45.968Z",
  "name": "Fixed line outage",
  "priority": "High",
  "requestedResolutionDate": "2019-05-31T07:34:45.968Z",
  "severity": "Major",
  "status": "pending",
  "statusChangeDate": "2019-05-31T07:34:45.968Z",
  "statusChangeReason": "Need more information from the customer",
  "ticketType": "Outage",
  "attachment": [
    {
      "id": "2384..",
      "mimeType": "image/png",
      "name": "CPE Diagnostic Output",
      "url": "https://mycsp.com/docloader?docnum=3534555",
      "@type": "AttachmentRef"
    }
  ],
  "channel": {
    "id": "4a62..",
    "name": "self service",
    "@type": "ChannelRef"
  },
  "note": [
    {
      "id": "5e74..",
      "author": "Jack Smith",
      "date": "2024-02-10T14:34:45.968Z",
      "@type": "Note"
    },
    {
      "id": "a7f2..",
      "author": "Emily Johnson",
      "date": "2024-02-01T11:34:45.968Z",
      "@type": "Note"
    }
  ]
}
```

```

        "id": "9b8a..",
        "author": "Support Team",
        "date": "2024-01-31T07:40:45.968Z",
        "text": "Urgent: We are currently experiencing a fixed line outage. Please
address this issue immediately.",
        "@type": "Note"
    }
],
"relatedParty": [
    {
        "role": "member",
        "partyOrPartyRole": {
            "id": "eb10..",
            "href": "{apiRoot}/partyManagement/v5/individual/eb10..",
            "name": "Sandy Smith",
            "@type": "PartyRef",
            "@referredType": "Individual"
        },
        "@type": "RelatedPartyRefOrPartyRoleRef"
    },
    {
        "role": "reporter",
        "partyOrPartyRole": {
            "id": "387d..",
            "name": "Jacob Jac Miller",
            "@type": "PartyRef",
            "@referredType": "Individual"
        },
        "@type": "RelatedPartyRefOrPartyRoleRef"
    },
    {
        "role": "customer",
        "partyOrPartyRole": {
            "id": "b811..",
            "href": "{apiRoot}/partyManagement/v5/customer/b811..",
            "name": "Jack Smith",
            "@type": "PartyRoleRef",
            "@referredType": "Customer"
        },
        "@type": "RelatedPartyRefOrPartyRoleRef"
    }
],
"@type": "TroubleTicket"
}

```

5.4. Removing a complete complex structure component of an array

Considering the resource example defined in [\[troubleTicketExample\]](#).

Remove from the `relatedParty` array, the element/structure identified by a child element condition, where the condition `name='Sandy Smith'` is true.

5.4.1. Using the JSON Path

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..  
Content-Type: application/json-patch-query+json
```

```
[  
  {  
    "op": "remove",  
    "path": "$.relatedParty[?(@.partyOrPartyRole.name=='Sandy Smith')]"  
  }  
]
```



The above JSON Path expression will resolve to `$['relatedParty'][0]`

5.4.2. Using JSON Path filter selector

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..  
Content-Type: application/json-patch-query+json
```

```
[  
  {  
    "op": "remove",  
    "path": "?filter=relatedParty[?(@.partyOrPartyRole.name=='Sandy Smith')]"  
  }  
]
```



The above JSON Path expression will resolve to `$['relatedParty'][0]`



As per [Design Guidelines – Part 6](#), for the **filter** selector the **\$** can be omitted.

All the above would result in the following updated Trouble Ticket resource:

RESPONSE

200 OK
Content-Type: application/json

```
{
  "id": "64f1..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/64f1..",
  "creationDate": "2024-01-31T07:34:45.968Z",
  "description": "Reporting a fixed line outage. Please investigate and resolve as soon as possible.",
  "expectedResolutionDate": "2019-06-10T07:34:45.968Z",
  "lastUpdate": "2024-02-10T07:34:45.968Z",
  "name": "Fixed line outage",
  "priority": "High",
  "requestedResolutionDate": "2019-05-31T07:34:45.968Z",
  "severity": "Major",
  "status": "pending",
  "statusChangeDate": "2019-05-31T07:34:45.968Z",
  "statusChangeReason": "Need more information from the customer",
  "ticketType": "Outage",
  "attachment": [
    {
      "id": "2384..",
      "attachmentType": "screenshot",
      "mimeType": "image/png",
      "name": "CPE Diagnostic Output",
      "url": "https://mycsp.com/docloader?docnum=3534555",
      "@type": "AttachmentRef"
    }
  ],
  "channel": {
    "id": "4a62..",
    "name": "self service",
    "@type": "ChannelRef"
  },
  "note": [
```

```

{
  "id": "5e74..",
  "author": "Jack Smith",
  "date": "2024-02-10T14:34:45.968Z",
  "@type": "Note"
},
{
  "id": "a7f2..",
  "author": "Emily Johnson",
  "date": "2024-02-01T11:34:45.968Z",
  "text": "Urgent matter, contact me at your earliest convenience",
  "@type": "Note"
},
{
  "id": "9b8a..",
  "author": "Support Team",
  "date": "2024-01-31T07:40:45.968Z",
  "text": "Urgent: We are currently experiencing a fixed line outage. Please
address this issue immediately.",
  "@type": "Note"
}
],
"relatedParty": [
  {
    "role": "reporter",
    "partyOrPartyRole": {
      "id": "387d..",
      "name": "Jacob Jac Miller",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  },
  {
    "role": "customer",
    "partyOrPartyRole": {
      "id": "b811..",
      "href": "{apiRoot}/partyManagement/v5/customer/b811..",
      "name": "Jack Smith",
      "@type": "PartyRoleRef",
      "@referredType": "Customer"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  }
]

```

```
"@type": "TroubleTicket"
}
```

5.5. Replacing an attribute from one of the components of an array

Considering the resource example defined in [\[troubleTicketExample\]](#).

Update the attribute named **name** in **attachment** structure for the component within the array whose attribute **id** has a value of **2384..**, the new value of **name** must be set to **CPE Product Test**

5.5.1. Using the JSON Path

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..
Content-Type: application/json-patch-query+json
```

```
[
  {
    "op": "replace",
    "path": "$.attachment[?(@.name=='CPE Diagnostic Output')].name",
    "value": "CPE Product Test"
  }
]
```



The above JSON Path expression will resolve to `$['attachment'][0]['name']`

5.5.2. Using JSON Path filter selector

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..
Content-Type: application/json-patch-query+json
```

```
[
  {
    "op": "replace",
```

```

    "path": "?filter=attachment[?(@.name=='CPE Diagnostic Output')].name",
    "value": "CPE Product Test"
  }
]

```



The above JSON Path expression will resolve to `$['attachment'][0]['name']`



As per [Design Guidelines – Part 6](#), for the **filter** selector the **\$** can be omitted.

All the above would result in the following updated Trouble Ticket resource:

```

{
  "id": "64f1..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/64f1..",
  "creationDate": "2024-01-31T07:34:45.968Z",
  "description": "Reporting a fixed line outage. Please investigate and resolve as soon as possible.",
  "expectedResolutionDate": "2019-06-10T07:34:45.968Z",
  "lastUpdate": "2024-02-10T07:34:45.968Z",
  "name": "Fixed line outage",
  "priority": "High",
  "requestedResolutionDate": "2019-05-31T07:34:45.968Z",
  "severity": "Major",
  "status": "pending",
  "statusChangeDate": "2019-05-31T07:34:45.968Z",
  "statusChangeReason": "Need more information from the customer",
  "ticketType": "Outage",
  "attachment": [
    {
      "id": "2384..",
      "attachmentType": "screenshot",
      "mimeType": "image/png",
      "name": "CPE Product Test",
      "url": "https://mycsp.com/docloader?docnum=3534555",
      "@type": "AttachmentRef"
    }
  ],
  "channel": {
    "id": "4a62..",
    "name": "self service",
    "@type": "ChannelRef"
  }
}

```

```

},
"note": [
  {
    "id": "5e74..",
    "author": "Jack Smith",
    "date": "2024-02-10T14:34:45.968Z",
    "@type": "Note"
  },
  {
    "id": "a7f2..",
    "author": "Emily Johnson",
    "date": "2024-02-01T11:34:45.968Z",
    "text": "Urgent matter, contact me at your earliest convenience",
    "@type": "Note"
  },
  {
    "id": "9b8a..",
    "author": "Support Team",
    "date": "2024-01-31T07:40:45.968Z",
    "text": "Urgent: We are currently experiencing a fixed line outage. Please address this issue immediately.",
    "@type": "Note"
  }
],
"relatedParty": [
  {
    "role": "member",
    "partyOrPartyRole": {
      "id": "eb10..",
      "href": "{apiRoot}/partyManagement/v5/individual/eb10..",
      "name": "Sandra Smith",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  },
  {
    "role": "reporter",
    "partyOrPartyRole": {
      "id": "387d..",
      "name": "Jacob Jac Miller",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  }
]

```

```

    },
    {
      "role": "customer",
      "partyOrPartyRole": {
        "id": "b811..",
        "href": "{apiRoot}/partyManagement/v5/customer/b811..",
        "name": "Jack Smith",
        "@type": "PartyRoleRef",
        "@referredType": "Customer"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    }
  ],
  "@type": "TroubleTicket"
}

```

5.6. Replacing a complete component of an array

Considering the resource example defined in [\[troubleTicketExample\]](#).

Update the **relatedParty** structure, the component within the array whose attribute **name** has a value of **Sandy Smith**, replacing the complete structure of that component with the new one provided.

5.6.1. Using the JSON Path

REQUEST

```

PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..
Content-Type: application/json-patch-query+json

```

```

[
  {
    "op": "replace",
    "path": "$.relatedParty[?(@.partyOrPartyRole.name=='Sandy Smith')].partyOrPartyRole",
    "value": {
      "id": "923e..",
      "name": "Jacob Jac Miller",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  }
]

```

```
}
]
```



The above JSON Path expression will resolve to `$['relatedParty'][0]['partyOrPartyRole']`

5.6.2. Using JSON Path filter selector

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..
Content-Type: application/json-patch+json
```

```
[
  {
    "op": "replace",
    "path": "?filter=relatedParty[?(@.partyOrPartyRole.name=='Sandy
Smith')].partyOrPartyRole",
    "value": {
      "id": "923e..",
      "name": "Jacob Jac Miller",
      "@type": "PartyRef",
      "@referredType": "Individual",
      "href": "{apiRoot}/partyManagement/v5/individual/923e.."
    }
  }
]
```



The above JSON Path expression will resolve to `$['relatedParty'][0]['partyOrPartyRole']`



As per [Design Guidelines – Part 6](#), for the **filter** selector the **\$** can be omitted.

All the above would result in the following updated Trouble Ticket resource:

```
{
  "id": "64f1..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/64f1..",
  "creationDate": "2024-01-31T07:34:45.968Z",
```

```
"description": "Reporting a fixed line outage. Please investigate and resolve
as soon as possible.",
"expectedResolutionDate": "2019-06-10T07:34:45.968Z",
"lastUpdate": "2024-02-10T07:34:45.968Z",
"name": "Fixed line outage",
"priority": "High",
"requestedResolutionDate": "2019-05-31T07:34:45.968Z",
"severity": "Major",
"status": "pending",
"statusChangeDate": "2019-05-31T07:34:45.968Z",
"statusChangeReason": "Need more information from the customer",
"ticketType": "Outage",
"attachment": [
  {
    "id": "2384..",
    "attachmentType": "screenshot",
    "mimeType": "image/png",
    "name": "CPE Diagnostic Output",
    "url": "https://mycsp.com/docloader?docnum=3534555",
    "@type": "AttachmentRef"
  }
],
"channel": {
  "id": "4a62..",
  "name": "self service",
  "@type": "ChannelRef"
},
"note": [
  {
    "id": "5e74..",
    "author": "Jack Smith",
    "date": "2024-02-10T14:34:45.968Z",
    "@type": "Note"
  },
  {
    "id": "a7f2..",
    "author": "Emily Johnson",
    "date": "2024-02-01T11:34:45.968Z",
    "text": "Urgent matter, contact me at your earliest convenience",
    "@type": "Note"
  },
  {
    "id": "9b8a..",
    "author": "Support Team",
    "date": "2024-01-31T07:40:45.968Z",
```

```

    "text": "Urgent: We are currently experiencing a fixed line outage. Please
address this issue immediately.",
    "@type": "Note"
  },
  "relatedParty": [
    {
      "role": "member",
      "partyOrPartyRole": {
        "id": "923e..",
        "name": "Jacob Jac Miller",
        "href": "{apiRoot}/partyManagement/v5/individual/923e..",
        "@type": "PartyRef",
        "@referredType": "Individual"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    },
    {
      "role": "reporter",
      "partyOrPartyRole": {
        "id": "923e..",
        "name": "Jacob Jac Miller",
        "@type": "PartyRef",
        "@referredType": "Individual"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    },
    {
      "role": "customer",
      "partyOrPartyRole": {
        "id": "b811..",
        "href": "{apiRoot}/partyManagement/v5/customer/b811..",
        "name": "Jack Smith",
        "@type": "PartyRoleRef",
        "@referredType": "Customer"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    }
  ],
  "@type": "TroubleTicket"
}

```

5.7. Replacing an attribute from one of the components of a complex array (resolving ambiguities)

Considering the resource example defined in [\[troubleTicketExample\]](#).

Update the **role** parameter in the **relatedParty** structure, for the component within the array whose attribute **@type=PartyRef** and **@referredType=Individual** and **name=Sandy Smith**, with the new value for **role=reporter**



Considering the example payload, it would be sufficient to pass the **name=Sandy Smith** as it is unique within the collection however the example is done to demonstrate multiple condition to resolve ambiguities.

5.7.1. Using the JSON Path

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..  
Content-Type: application/json-patch-query+json
```

```
[  
  {  
    "op": "replace",  
    "path": "$.relatedParty[?(@.partyOrPartyRole['@type']=='PartyRef' &&  
@.partyOrPartyRole['@referredType']=='Individual' &&  
@.partyOrPartyRole['name']=='Sandy Smith')].role",  
    "value": "reporter"  
  }  
]
```



The above JSON Path expression will resolve to **"\$['relatedParty'][0]['role']"**

5.7.2. Using the query expression with filter selector

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..  
Content-Type: application/json-patch-query+json
```

```
[
  {
    "op": "replace",
    "path": "?filter=relatedParty[?(@.partyOrPartyRole['@type']=='PartyRef' &&
    @.partyOrPartyRole['@referredType']=='Individual' &&
    @.partyOrPartyRole['name']=='Sandy Smith')].role",
    "value": "reporter"
  }
]
```



As per [Design Guidelines – Part 6](#), for the **filter** selector the **\$** can be omitted.

All the above would result in the following updated Trouble Ticket resource:

```
{
  "id": "64f1..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/64f1..",
  "creationDate": "2024-01-31T07:34:45.968Z",
  "description": "Reporting a fixed line outage. Please investigate and resolve
as soon as possible.",
  "expectedResolutionDate": "2019-06-10T07:34:45.968Z",
  "lastUpdate": "2024-02-10T07:34:45.968Z",
  "name": "Fixed line outage",
  "priority": "High",
  "requestedResolutionDate": "2019-05-31T07:34:45.968Z",
  "severity": "Major",
  "status": "pending",
  "statusChangeDate": "2019-05-31T07:34:45.968Z",
  "statusChangeReason": "Need more information from the customer",
  "ticketType": "Outage",
  "attachment": [
    {
      "id": "2384..",
      "attachmentType": "screenshot",
      "mimeType": "image/png",
      "name": "CPE Diagnostic Output",
      "url": "https://mycsp.com/docloader?docnum=3534555",
      "@type": "AttachmentRef"
    }
  ],
  "channel": {
```

```
"id": "4a62..",
"name": "self service",
"@type": "ChannelRef"
},
"note": [
  {
    "id": "5e74..",
    "author": "Jack Smith",
    "date": "2024-02-10T14:34:45.968Z",
    "@type": "Note"
  },
  {
    "id": "a7f2..",
    "author": "Emily Johnson",
    "date": "2024-02-01T11:34:45.968Z",
    "text": "Urgent matter, contact me at your earliest convenience",
    "@type": "Note"
  },
  {
    "id": "9b8a..",
    "author": "Support Team",
    "date": "2024-01-31T07:40:45.968Z",
    "text": "Urgent: We are currently experiencing a fixed line outage. Please address this issue immediately.",
    "@type": "Note"
  }
],
"relatedParty": [
  {
    "role": "reporter",
    "partyOrPartyRole": {
      "id": "eb10..",
      "href": "{apiRoot}/partyManagement/v5/individual/eb10..",
      "name": "Sandy Smith",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  },
  {
    "role": "reporter",
    "partyOrPartyRole": {
      "id": "387d..",
      "name": "Jacob Jac Miller",
      "@type": "PartyRef",
```

```
    "@referredType": "Individual"
  },
  "@type": "RelatedPartyRefOrPartyRoleRef"
},
{
  "role": "customer",
  "partyOrPartyRole": {
    "id": "b811..",
    "href": "{apiRoot}/partyManagement/v5/customer/b811..",
    "name": "Jack Smith",
    "@type": "PartyRoleRef",
    "@referredType": "Customer"
  },
  "@type": "RelatedPartyRefOrPartyRoleRef"
}
],
"@type": "TroubleTicket"
}
```

Chapter 6. Query Expression with **.(dot)** notation

The following is an example JSON Patch Query document, transferred in an HTTP PATCH request, where the path query expression has been used:

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..
Content-Type: application/json-patch-query+json
```

```
[
  {"op": "remove", "path": "/a/b/c?elemInC=value" },
  {"op": "replace", "path": "/a/b/c?elemInB=value", "value": 42}
]
```

The **path** definition includes the query expression and has the following structure and components: **<element> ? <condition>**, where:

- **<element>**
 - represents the resource component that is subject to the operation
- **?**
 - represents the character that delimit the two components
- **<condition>**
 - represents the condition that can be applied to the resource and MUST resolve to TRUE to identify the correct element
 - the condition can be specified either using the **.(dot)** notation described in [\[TMF630-1\]](#), section Complex attribute filtering using the **.(dot)** notation.

This section presents a set of examples of different PATCH update operations using JSON Patch Query document.

6.1. Adding an attribute to one of the components from an array

Considering the resource example defined in [\[troubleTicketExample\]](#).

Add to the **attachment** array, for the element where the condition **id=2384..** evaluates to true, a new attribute **description** with a value of **CPE Output**

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..  
Content-Type: application/json-patch-query+json
```

```
[  
  {  
    "op": "add",  
    "path": "/attachment/description?attachment.id=2384..",  
    "value": "CPE Output"  
  }  
]
```

The path expression used, `/attachment/description?attachment.id=2384..`:

- `/attachment/description` – path and name of the element to be added
- `?` – gen-delim
- `attachment.id=2384..` – condition which must be evaluated to *true*

The above would result in the following updated Trouble Ticket resource:

RESPONSE

```
200 OK  
Content-Type: application/json
```

```
{  
  "id": "64f1..",  
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/64f1..",  
  "creationDate": "2024-01-31T07:34:45.968Z",  
  "description": "Reporting a fixed line outage. Please investigate and resolve  
as soon as possible.",  
  "expectedResolutionDate": "2019-06-10T07:34:45.968Z",  
  "lastUpdate": "2024-02-10T07:34:45.968Z",  
  "name": "Fixed line outage",  
  "priority": "High",  
  "requestedResolutionDate": "2019-05-31T07:34:45.968Z",  
  "severity": "Major",  
  "status": "pending",  
  "statusChangeDate": "2019-05-31T07:34:45.968Z",  
}
```

```
"statusChangeReason": "Need more information from the customer",
"ticketType": "Outage",
"attachment": [
  {
    "id": "2384..",
    "attachmentType": "screenshot",
    "description": "CPE Output",
    "mimeType": "image/png",
    "name": "CPE Diagnostic Output",
    "url": "https://mycsp.com/docloader?docnum=3534555",
    "@type": "AttachmentRef"
  }
],
"channel": {
  "id": "4a62..",
  "name": "self service",
  "@type": "ChannelRef"
},
"note": [
  {
    "id": "5e74..",
    "author": "Jack Smith",
    "date": "2024-02-10T14:34:45.968Z",
    "text": "This is quite important, please approach me as soon as possible",
    "@type": "Note"
  },
  {
    "id": "a7f2..",
    "author": "Emily Johnson",
    "date": "2024-02-01T11:34:45.968Z",
    "text": "Urgent matter, contact me at your earliest convenience",
    "@type": "Note"
  },
  {
    "id": "9b8a..",
    "author": "Support Team",
    "date": "2024-01-31T07:40:45.968Z",
    "text": "Urgent: We are currently experiencing a fixed line outage. Please address this issue immediately.",
    "@type": "Note"
  }
],
"@type": "TroubleTicket"
}
```

6.2. Removing one of the components of an Array Element (the whole structure)

Considering the resource example defined in [\[troubleTicketExample\]](#).

Remove from the **note** array, identified by the path provided (**/note**), the element/structure whose attribute **author** has a value of **Jack Smith**.

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..  
Content-Type: application/json-patch-query+json
```

```
[  
  {  
    "op": "remove",  
    "path": "/note?note.author=Jack Smith"  
  }  
]
```

The path expression used, **/note?note.author='Jack Smith'**:

- **/note** – path identifying the array whose elements should be removed
- **?** – gen-delim
- **note.author='Jack Smith'** – condition which must be evaluated to *true*

All the above would result in the following updated Trouble Ticket resource:

RESPONSE

```
200 OK  
Content-Type: application/json
```

```
{  
  "id": "64f1..",  
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/64f1..",  
  "creationDate": "2024-01-31T07:34:45.968Z",  
  "description": "Reporting a fixed line outage. Please investigate and resolve  
as soon as possible.",  
  "expectedResolutionDate": "2019-06-10T07:34:45.968Z",  
}
```

```
"lastUpdate": "2024-02-10T07:34:45.968Z",
"name": "Fixed line outage",
"priority": "High",
"requestedResolutionDate": "2019-05-31T07:34:45.968Z",
"severity": "Major",
"status": "pending",
"statusChangeDate": "2019-05-31T07:34:45.968Z",
"statusChangeReason": "Need more information from the customer",
"ticketType": "Outage",
"attachment": [
  {
    "id": "2384..",
    "attachmentType": "screenshot",
    "mimeType": "image/png",
    "name": "CPE Diagnostic Output",
    "url": "https://mycsp.com/docloader?docnum=3534555",
    "@type": "AttachmentRef"
  }
],
"channel": {
  "id": "4a62..",
  "name": "self service",
  "@type": "ChannelRef"
},
"note": [
  {
    "id": "a7f2..",
    "author": "Emily Johnson",
    "date": "2024-02-01T11:34:45.968Z",
    "text": "Urgent matter, contact me at your earliest convenience",
    "@type": "Note"
  },
  {
    "id": "9b8a..",
    "author": "Support Team",
    "date": "2024-01-31T07:40:45.968Z",
    "text": "Urgent: We are currently experiencing a fixed line outage. Please address this issue immediately.",
    "@type": "Note"
  }
],
"relatedParty": [
  {
    "role": "member",
    "partyOrPartyRole": {
```

```

        "id": "eb10..",
        "href": "{apiRoot}/partyManagement/v5/individual/eb10..",
        "name": "Sandy Smith",
        "@type": "PartyRef",
        "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
},
{
    "role": "reporter",
    "partyOrPartyRole": {
        "id": "387d..",
        "name": "Jacob Jac Miller",
        "@type": "PartyRef",
        "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
},
{
    "role": "customer",
    "partyOrPartyRole": {
        "id": "b811..",
        "href": "{apiRoot}/partyManagement/v5/customer/b811..",
        "name": "Jack Smith",
        "@type": "PartyRoleRef",
        "@referredType": "Customer"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
}
],
"@type": "TroubleTicket"
}

```

6.3. Removing an attribute from one of the components of an array

Considering the resource example defined in [\[troubleTicketExample\]](#).

Remove from the **attachment** array, the attribute from the element/structure identified by the path provided (**/attachment/attachmentType**), where the condition **id=2384..** is true

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..
```

```
Content-Type: application/json-patch-query+json
```

```
[
  {
    "op": "remove",
    "path": "/attachment/attachmentType?attachment.id=2384.."
  }
]
```

The path expression used, `/attachment/attachmentType?attachment.id='2384..'`:

- `/attachment/attachmentType` – path identifying the element that should be removed
- `?` – gen-delim
- `attachment.id=2384..` – condition which must be evaluated to *true*

All the above would result in the following updated Trouble Ticket resource:

RESPONSE

```
200 OK
Content-Type: application/json
```

```
{
  "id": "64f1..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/64f1..",
  "creationDate": "2024-01-31T07:34:45.968Z",
  "description": "Reporting a fixed line outage. Please investigate and resolve as soon as possible.",
  "expectedResolutionDate": "2019-06-10T07:34:45.968Z",
  "lastUpdate": "2024-02-10T07:34:45.968Z",
  "name": "Fixed line outage",
  "priority": "High",
  "requestedResolutionDate": "2019-05-31T07:34:45.968Z",
  "severity": "Major",
  "status": "pending",
  "statusChangeDate": "2019-05-31T07:34:45.968Z",
  "statusChangeReason": "Need more information from the customer",
  "ticketType": "Outage",
  "attachment": [
```

```

{
  "id": "2384..",
  "mimeType": "image/png",
  "name": "CPE Diagnostic Output",
  "url": "https://mycsp.com/docloader?docnum=3534555",
  "@type": "AttachmentRef"
},
{
  "id": "4a62..",
  "name": "self service",
  "@type": "ChannelRef"
},
{
  "note": [
    {
      "id": "5e74..",
      "author": "Jack Smith",
      "date": "2024-02-10T14:34:45.968Z",
      "@type": "Note"
    },
    {
      "id": "a7f2..",
      "author": "Emily Johnson",
      "date": "2024-02-01T11:34:45.968Z",
      "@type": "Note"
    },
    {
      "id": "9b8a..",
      "author": "Support Team",
      "date": "2024-01-31T07:40:45.968Z",
      "text": "Urgent: We are currently experiencing a fixed line outage. Please address this issue immediately.",
      "@type": "Note"
    }
  ],
  "relatedParty": [
    {
      "role": "member",
      "partyOrPartyRole": {
        "id": "eb10..",
        "href": "{apiRoot}/partyManagement/v5/individual/eb10..",
        "name": "Sandy Smith",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],

```

```

    "@type": "RelatedPartyRefOrPartyRoleRef"
  },
  {
    "role": "reporter",
    "partyOrPartyRole": {
      "id": "387d..",
      "name": "Jacob Jac Miller",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  },
  {
    "role": "customer",
    "partyOrPartyRole": {
      "id": "b811..",
      "href": "{apiRoot}/partyManagement/v5/customer/b811..",
      "name": "Jack Smith",
      "@type": "PartyRoleRef",
      "@referredType": "Customer"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  }
],
"@type": "TroubleTicket"
}

```

6.4. Removing a complete complex structure component of an array

Considering the resource example defined in [\[troubleTicketExample\]](#).

Remove from the **relatedParty** array, the element/structure identified by a child element condition, where the condition **name='Sandy Smith'** is true.

REQUEST

```

PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..
Content-Type: application/json-patch-query+json

```

```

[
  {
    "op": "remove",

```

```

    "path": "/relatedParty?partyOrPartyRole.name=Sandy Smith"
  }
]

```

The path expression used, `/relatedParty?partyOrPartyRole.name='Sandy Smith'`:

- `/relatedParty` – path identifying the array
- `?` – gen-delim
- `partyOrPartyRole.name='Sandy Smith'` – condition which must be evaluated to *true*

All the above would result in the following updated Trouble Ticket resource:

RESPONSE

```

200 OK
Content-Type: application/json

```

```

{
  "id": "64f1..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/64f1..",
  "creationDate": "2024-01-31T07:34:45.968Z",
  "description": "Reporting a fixed line outage. Please investigate and resolve
as soon as possible.",
  "expectedResolutionDate": "2019-06-10T07:34:45.968Z",
  "lastUpdate": "2024-02-10T07:34:45.968Z",
  "name": "Fixed line outage",
  "priority": "High",
  "requestedResolutionDate": "2019-05-31T07:34:45.968Z",
  "severity": "Major",
  "status": "pending",
  "statusChangeDate": "2019-05-31T07:34:45.968Z",
  "statusChangeReason": "Need more information from the customer",
  "ticketType": "Outage",
  "attachment": [
    {
      "id": "2384..",
      "attachmentType": "screenshot",
      "mimeType": "image/png",
      "name": "CPE Diagnostic Output",
      "url": "https://mycsp.com/docloader?docnum=3534555",
      "@type": "AttachmentRef"
    }
  ]
}

```

```
],
"channel": {
  "id": "4a62..",
  "name": "self service",
  "@type": "ChannelRef"
},
"note": [
  {
    "id": "5e74..",
    "author": "Jack Smith",
    "date": "2024-02-10T14:34:45.968Z",
    "@type": "Note"
  },
  {
    "id": "a7f2..",
    "author": "Emily Johnson",
    "date": "2024-02-01T11:34:45.968Z",
    "text": "Urgent matter, contact me at your earliest convenience",
    "@type": "Note"
  },
  {
    "id": "9b8a..",
    "author": "Support Team",
    "date": "2024-01-31T07:40:45.968Z",
    "text": "Urgent: We are currently experiencing a fixed line outage. Please address this issue immediately.",
    "@type": "Note"
  }
],
"relatedParty": [
  {
    "role": "reporter",
    "partyOrPartyRole": {
      "id": "387d..",
      "name": "Jacob Jac Miller",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  },
  {
    "role": "customer",
    "partyOrPartyRole": {
      "id": "b811..",
      "href": "{apiRoot}/partyManagement/v5/customer/b811..",

```

```

      "name": "Jack Smith",
      "@type": "PartyRoleRef",
      "@referredType": "Customer"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  }
],
"@type": "TroubleTicket"
}

```

6.5. Replacing an attribute from one of the components of an array

Considering the resource example defined in [\[troubleTicketExample\]](#).

Update the attribute named **name** in **attachment.name** structure for the component within the array whose attribute **id** has a value of **2384..**, the new value of **name** must be set to **CPE Product Test**

REQUEST

```

PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..
Content-Type: application/json-patch-query+json

```

```

[
  {
    "op": "replace",
    "path": "/attachment/description?attachment.description=CPE Diagnostic
Output",
    "value": "CPE Product Test"
  }
]

```

The path expression used, **/attachment/description?attachment.description='CPE Diagnostic Output'**:

- **/attachment/description** – path identifying the element whose value will be updated
- **?** – gen-delim
- **attachment.description='CPE Diagnostic Output'** – condition which must be evaluated to **true**

All the above would result in the following updated Trouble Ticket resource:

```
{
  "id": "64f1..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/64f1..",
  "creationDate": "2024-01-31T07:34:45.968Z",
  "description": "Reporting a fixed line outage. Please investigate and resolve
as soon as possible.",
  "expectedResolutionDate": "2019-06-10T07:34:45.968Z",
  "lastUpdate": "2024-02-10T07:34:45.968Z",
  "name": "Fixed line outage",
  "priority": "High",
  "requestedResolutionDate": "2019-05-31T07:34:45.968Z",
  "severity": "Major",
  "status": "pending",
  "statusChangeDate": "2019-05-31T07:34:45.968Z",
  "statusChangeReason": "Need more information from the customer",
  "ticketType": "Outage",
  "attachment": [
    {
      "id": "2384..",
      "attachmentType": "screenshot",
      "mimeType": "image/png",
      "name": "CPE Product Test",
      "url": "https://mycsp.com/docloader?docnum=3534555",
      "@type": "AttachmentRef"
    }
  ],
  "channel": {
    "id": "4a62..",
    "name": "self service",
    "@type": "ChannelRef"
  },
  "note": [
    {
      "id": "5e74..",
      "author": "Jack Smith",
      "date": "2024-02-10T14:34:45.968Z",
      "@type": "Note"
    },
    {
      "id": "a7f2..",
      "author": "Emily Johnson",
      "date": "2024-02-01T11:34:45.968Z",
      "text": "Urgent matter, contact me at your earliest convenience",
      "@type": "Note"
    }
  ]
}
```

```
    },
    {
      "id": "9b8a..",
      "author": "Support Team",
      "date": "2024-01-31T07:40:45.968Z",
      "text": "Urgent: We are currently experiencing a fixed line outage. Please address this issue immediately.",
      "@type": "Note"
    }
  ],
  "relatedParty": [
    {
      "role": "member",
      "partyOrPartyRole": {
        "id": "eb10..",
        "href": "{apiRoot}/partyManagement/v5/individual/eb10..",
        "name": "Sandra Smith",
        "@type": "PartyRef",
        "@referredType": "Individual"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    },
    {
      "role": "reporter",
      "partyOrPartyRole": {
        "id": "387d..",
        "name": "Jacob Jac Miller",
        "@type": "PartyRef",
        "@referredType": "Individual"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    },
    {
      "role": "customer",
      "partyOrPartyRole": {
        "id": "b811..",
        "href": "{apiRoot}/partyManagement/v5/customer/b811..",
        "name": "Jack Smith",
        "@type": "PartyRoleRef",
        "@referredType": "Customer"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    }
  ],
  "@type": "TroubleTicket"
```

```
}
```

6.6. Replacing a complete component of an array

Considering the resource example defined in [\[troubleTicketExample\]](#).

Update the **relatedParty** structure, the component within the array whose attribute **name** has a value of **Sandy Smith**, replacing the complete structure of that component with the new one provided.

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..
Content-Type: application/json-patch-query+json
```

```
[
  {
    "op": "replace",
    "path":
"/relatedParty/partyOrPartyRole?relatedParty.partyOrPartyRole.name=Sandy Smith",
    "value": {
      "id": "923e..",
      "name": "Jacob Jac Miller",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  }
]
```

The path expression used, **/relatedParty/partyOrPartyRole?relatedParty.role='reporter'**:

- **/relatedParty/partyOrPartyRole** – path identifying the element whose value will be updated
- **?** – gen-delim
- **relatedParty.role='reporter'** – condition which must be evaluated to *true*

All the above would result in the following updated Trouble Ticket resource:

```
{
  "id": "64f1..",
```

```
"href": "{apiRoot}/troubleTicket/v5/troubleTicket/64f1..",
"creationDate": "2024-01-31T07:34:45.968Z",
"description": "Reporting a fixed line outage. Please investigate and resolve
as soon as possible.",
"expectedResolutionDate": "2019-06-10T07:34:45.968Z",
"lastUpdate": "2024-02-10T07:34:45.968Z",
"name": "Fixed line outage",
"priority": "High",
"requestedResolutionDate": "2019-05-31T07:34:45.968Z",
"severity": "Major",
"status": "pending",
"statusChangeDate": "2019-05-31T07:34:45.968Z",
"statusChangeReason": "Need more information from the customer",
"ticketType": "Outage",
"attachment": [
  {
    "id": "2384..",
    "attachmentType": "screenshot",
    "mimeType": "image/png",
    "name": "CPE Diagnostic Output",
    "url": "https://mycsp.com/docloader?docnum=3534555",
    "@type": "AttachmentRef"
  }
],
"channel": {
  "id": "4a62..",
  "name": "self service",
  "@type": "ChannelRef"
},
"note": [
  {
    "id": "5e74..",
    "author": "Jack Smith",
    "date": "2024-02-10T14:34:45.968Z",
    "@type": "Note"
  },
  {
    "id": "a7f2..",
    "author": "Emily Johnson",
    "date": "2024-02-01T11:34:45.968Z",
    "text": "Urgent matter, contact me at your earliest convenience",
    "@type": "Note"
  },
  {
    "id": "9b8a..",
```

```

    "author": "Support Team",
    "date": "2024-01-31T07:40:45.968Z",
    "text": "Urgent: We are currently experiencing a fixed line outage. Please
address this issue immediately.",
    "@type": "Note"
  }
],
"relatedParty": [
  {
    "role": "member",
    "partyOrPartyRole": {
      "id": "eb10..",
      "href": "{apiRoot}/partyManagement/v5/individual/eb10..",
      "name": "Sandy Smith",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  },
  {
    "role": "reporter",
    "partyOrPartyRole": {
      "id": "923e..",
      "name": "Jacob Jac Miller",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  },
  {
    "role": "customer",
    "partyOrPartyRole": {
      "id": "b811..",
      "href": "{apiRoot}/partyManagement/v5/customer/b811..",
      "name": "Jack Smith",
      "@type": "PartyRoleRef",
      "@referredType": "Customer"
    },
    "@type": "RelatedPartyRefOrPartyRoleRef"
  }
],
"@type": "TroubleTicket"
}

```

6.7. Replacing an attribute from one of the components of a complex array (resolving ambiguities)

Considering the resource example defined in [\[troubleTicketExample\]](#).

Update the **role** parameter in the **relatedParty** structure, for the component within the array whose attribute **@type=PartyRef** and **@referredType=Individual** and **name=Sandy Smith**, with the new value for **role=reporter**



Considering the example payload, it would be suffice to pass the **name=Sandy Smith** as it is unique within the collection however the example is done to demonstrate multiple condition to resolve ambiguities.

REQUEST

```
PATCH {apiRoot}/troubleTicket/v5/troubleTicket/64f1..
Content-Type: application/json-patch-query+json
```

```
[
  {
    "op": "replace",
    "path":
"/relatedParty/role?relatedParty.partyOrPartyRole.@type=PartyRef&relatedParty.partyOrPartyRole.@referredType=Individual&relatedParty.partyOrPartyRole.name=Sandy Smith",
    "value": "reporter"
  }
]
```

The **path** expression used, **/relatedParty/role?relatedParty.partyOrPartyRole.@type=PartyRef&relatedParty.partyOrPartyRole.@referredType=Individual&relatedParty.partyOrPartyRole.name='Sandy Smith'**:

- **/relatedParty/role** – path identifying the element whose value will be updated
- **?** – gen-delim
- **relatedParty.partyOrPartyRole.@type=PartyRef&relatedParty.partyOrPartyRole.@referredType=Individual&relatedParty.partyOrPartyRole.name='Sandy Smith'** – condition which must be evaluated to *true*;

All the above would result in the following updated Trouble Ticket resource:

```
{
  "id": "64f1..",
  "href": "{apiRoot}/troubleTicket/v5/troubleTicket/64f1..",
  "creationDate": "2024-01-31T07:34:45.968Z",
  "description": "Reporting a fixed line outage. Please investigate and resolve
as soon as possible.",
  "expectedResolutionDate": "2019-06-10T07:34:45.968Z",
  "lastUpdate": "2024-02-10T07:34:45.968Z",
  "name": "Fixed line outage",
  "priority": "High",
  "requestedResolutionDate": "2019-05-31T07:34:45.968Z",
  "severity": "Major",
  "status": "pending",
  "statusChangeDate": "2019-05-31T07:34:45.968Z",
  "statusChangeReason": "Need more information from the customer",
  "ticketType": "Outage",
  "attachment": [
    {
      "id": "2384..",
      "attachmentType": "screenshot",
      "mimeType": "image/png",
      "name": "CPE Diagnostic Output",
      "url": "https://mycsp.com/docloader?docnum=3534555",
      "@type": "AttachmentRef"
    }
  ],
  "channel": {
    "id": "4a62..",
    "name": "self service",
    "@type": "ChannelRef"
  },
  "note": [
    {
      "id": "5e74..",
      "author": "Jack Smith",
      "date": "2024-02-10T14:34:45.968Z",
      "@type": "Note"
    },
    {
      "id": "a7f2..",
      "author": "Emily Johnson",
      "date": "2024-02-01T11:34:45.968Z",
      "text": "Urgent matter, contact me at your earliest convenience",
      "@type": "Note"
    }
  ]
}
```

```

    },
    {
      "id": "9b8a..",
      "author": "Support Team",
      "date": "2024-01-31T07:40:45.968Z",
      "text": "Urgent: We are currently experiencing a fixed line outage. Please address this issue immediately.",
      "@type": "Note"
    }
  ],
  "relatedParty": [
    {
      "role": "reporter",
      "partyOrPartyRole": {
        "id": "eb10..",
        "href": "{apiRoot}/partyManagement/v5/individual/eb10..",
        "name": "Sandy Smith",
        "@type": "PartyRef",
        "@referredType": "Individual"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    },
    {
      "role": "reporter",
      "partyOrPartyRole": {
        "id": "387d..",
        "name": "Jacob Jac Miller",
        "@type": "PartyRef",
        "@referredType": "Individual"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    },
    {
      "role": "customer",
      "partyOrPartyRole": {
        "id": "b811..",
        "href": "{apiRoot}/partyManagement/v5/customer/b811..",
        "name": "Jack Smith",
        "@type": "PartyRoleRef",
        "@referredType": "Customer"
      },
      "@type": "RelatedPartyRefOrPartyRoleRef"
    }
  ],
  "@type": "TroubleTicket"

```

}

References

- [RFC3986]
Uniform Resource Identifier (URI): Generic Syntax
IETF (Internet Engineering Task Force)
<https://www.ietf.org/rfc/rfc3986.html>
- [RFC4627]
JavaScript Object Notation (JSON)
IETF (Internet Engineering Task Force)
<https://datatracker.ietf.org/doc/html/rfc4627>
- [RFC5789]
PATCH Method for HTTP
IETF (Internet Engineering Task Force)
<https://www.ietf.org/rfc/rfc5789.html>
- [RFC6901]
JavaScript Object Notation (JSON) Pointer
IETF (Internet Engineering Task Force)
<https://www.ietf.org/rfc/rfc6901.html>
- [RFC6902]
JavaScript Object Notation (JSON) Patch
IETF (Internet Engineering Task Force)
<https://www.ietf.org/rfc/rfc6902.html>
- [RFC8174]
Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words
IETF (Internet Engineering Task Force)
<https://www.ietf.org/rfc/rfc8174.html>
- [RFC9110]
HTTP Semantics
IETF (Internet Engineering Task Force)
<https://www.ietf.org/rfc/rfc9110.html>
- [RFC9535]
JSONPath – Query Expressions for JSON
IETF (Internet Engineering Task Force)
<https://www.ietf.org/rfc/rfc9535.html>
- [TMF630-1]
TMForum
REST API Design Guidelines
Part 1: Practical guidelines for RESTful APIs naming, CRUD, filtering, tasks, history,

notifications

□ [TMF630-2]

TMForum

REST API Design Guidelines

Part 2: Advanced guidelines for RESTful APIs polymorphism, extension patterns, depth and expand directive, entity RefOrValue

□ [TMF630-3]

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REST API Design Guidelines

Part 3: Guidelines for extending TMF Open API's with hypermedia support

□ [TMF630-4]

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REST API Design Guidelines

Part 4: Advanced guidelines for import and export tasks and for entity versioning

□ [TMF630-5]

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REST API Design Guidelines

Part 5: JSON patch extensions to manage arrays

□ [TMF630-6]

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REST API Design Guidelines

Part 6: JSON path extensions

□ [TMF630-7]

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REST API Design Guidelines

Part 7: JSON Schemas guideline

Administrative Appendix

This Appendix provides additional background material about the TM Forum and this document. In general, sections may be included or omitted as desired; however, a Document History must always be included.

Document History

Version History

This section records the changes between this and the previous document version as it is edited by the team concerned. This is an incremental number which does not have to match the release number and used for change control purposes only.

Version Number	Date Modified	Modified by:	Description of changes
1.0	23-Nov-2014	Pierre Gauthier TM Forum	Description e.g. first issue of document
1.1.1	12/03/2015	Alicja Kawecki, TM Forum	Updated cover, footer and Notice to reflect TM Forum Approved status
2.0.0	08-Nov-2017	Peter Norbury	Update to Notice and change document number
2.0.1	12-Dec-2017	Adrienne Walcott	Formatting/style edits prior to publishing
2.0.2	20-Mar-2018	Adrienne Walcott	Updated to reflect TM Forum Approved Status
5.0.0	31-Apr-2024	Several Contributors	Expanded the support for the JSON Path in JSON Patch Query; Added further clarifications on the JSON Patch Query based on .(dot) notation

Release History

This section records the changes between this and the previous Official document release. The release number is the 'Marketing' number which this version of the document is first being assigned to.

Release Number	Date Modified	Modified by:	Description of changes
17.5.0	08-Nov-2017	Peter Norbury	First release of document
17.5.1	20-Mar-2018	Adrienne Walcott	Updated to reflect TM Forum Approved Status

Acknowledgments

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